

AuraRetail

THE OFFLINE-FIRST VOICE AI RETAIL CONTROL LAYER

A Unified Next.js Analytics Portal Bridging Familiar Cloud Sheets with Natural Speech Interfaces

OFFLINE-FIRST

VOICE AGENT AI

GOOGLE SHEETS CORE

- Why this project stands out:
- **Zero-overhead database setup:** Google Sheets is your production database.
 - **Voice-controlled operations:** Hands-free database query, sync, and client-side navigation.
 - **Fail-safe offline caching:** Seamlessly switches to local mock DB during connectivity dropouts.

PROJECT OVERVIEW & CORE PILLARS

1. THE REVOLUTIONARY PROBLEM

TARGET USER PAIN POINT

Small store owners are trapped between complex, expensive ERP software they do not understand and messy, paper logbooks that get damaged and fail to calculate margins or expiry warnings.

2. VOICE COMMAND AUTOMATION

INTELLIGENT UI PIPELINE

Captures browser microphone speech, transcribes, and forwards it to n8n AI workflows. Synthesizes responses using Google Translate TTS backend to speak directly to the shopkeeper.

3. MULTI-LEVEL RESILIENCE

BULLETPROOF OFFLINE CACHING

No network? No credentials? No problem. The dashboard instantly detects Sheets connection offline status, shifting all operations to a local cache ledger file without interrupting business flow.

4. SAFE TRANSACTION LEDGER

DATABASE MANAGEMENT ENGINE

Inline cell editing, batched changes buffering, and a single-row editing safety lock prevent database race-conditions. Auto-populates item specifications and prices to bypass manual errors.

AUDIENCE: HACKATHON EVALUATION PANEL

TECH STACK: NEXT.JS, GOOGLE APIS, N8N, TTS

RATING: TARGET GRAND PRIZE

1. The Real-World Problem & Solution Pitch

THE PROBLEM WE SOLVE: RETAIL MANAGEMENT IS BROKEN

Local mom-and-pop shops and small retail stores handle dozens of transactions daily. However, they face a critical dilemma:

1. Paper Logbooks are inefficient, easily lost or destroyed, and provide zero calculations of profitability, margin percentages, or stock valuation.
2. Traditional Spreadsheets are too complex, look cluttered on small screens, and are prone to data-entry mistakes (like entering letters in a price column or mismatched unit descriptions).
3. Enterprise ERP Systems are prohibitively expensive, require extensive training, and need massive processing power.
4. Product Expiry & Low Stock lead to massive waste and lost sales because shopkeepers do not notice items are expiring or running low on stock.

THE SOLUTION: AURARETAIL DASHBOARD

AuraRetail solves this by offering a lightweight, beautiful business intelligence portal that connects directly to Google Sheets (which shop owners already know and trust) and equips them with an interactive voice agent.

- Low Cost & Zero Overhead: Uses Google Sheets as a free, scalable cloud database. No SQL servers, hosting fees, or complicated logins.
- Voice Assistant Core: Shopkeepers can query sales, check stock, navigation pages, or sync spreadsheet records by simply speaking to the dashboard. The system speaks the responses back naturally.
- Bulletproof Reliability: Real-time background checking shifts database transactions to a local cache file (lib/mock_db.json) automatically if Google Sheets APIs are offline. Operations continue uninterrupted.

WHY THIS PROJECT IS UNIQUE & WINS HACKATHONS

- **Familiarity-Driven Database:** Instead of trying to teach operators a new, complex software, we utilize their existing Google Sheet. They edit cells in the cloud, and our app compiles the data. Changes sync instantly both ways.
- **Single-Row Modification Lock:** Prevents conflicting data entries. If a user is currently modifying values on row 5, the app locks edit access on all other rows, protecting the Google Sheets API from cell-level collision errors.
- **Dual Speech Engine Pipeline:** Integrates browser Webkit SpeechRecognition with n8n AI workflows. If n8n is offline, a local regex-based keyword search matches intents locally, reading KPIs directly from the active client state.
- **Zero-Config Server-Side TTS:** If n8n returns a text reply, our backend automatically proxies it to Google Translate API, generating a binary audio stream. The browser plays it directly, creating an interactive "talking" assistant.

2. Analytics Overview & Live KPI Metrics

The AuraRetail dashboard aggregates rows from the inventory and sales spreadsheets, computing metrics dynamically to display overall business health at a glance:

- **Total Stock Value:** Multiplies item quantities by their purchase cost to determine the total capital locked on store shelves.
- **Potential Revenue:** Aggregates stock valuation at active retail selling prices to show gross revenue potential.
- **Today's Profit & Revenue:** Tracks completed transactions for the current date, subtracting product cost basis to calculate net profit margins.
- **Average Profit Markup:** Calculates markups across all stock rows, giving operators insight into price optimizations.

VISUAL COMPILER & COMPREHENSIVE CHARTS

AuraRetail integrates Recharts to map sales histories: a Revenue Trend Chart (smooth linear-gradient area curves of sales volume) and a Best-Sellers Bar Chart (ranks products by units sold or gross earnings).

PROJECT SCREENSHOT: REAL-TIME ANALYTICS OVERVIEW

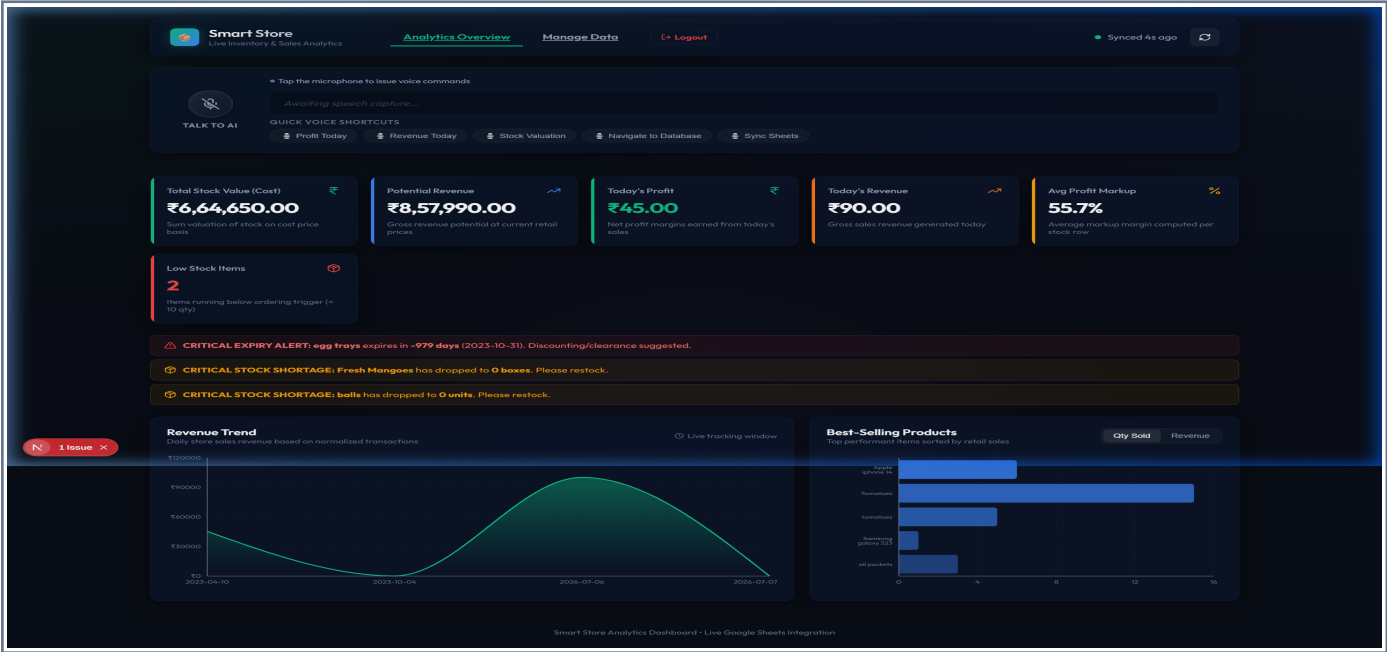


Figure 1: UI showing live KPI metrics, dynamic red/amber expiry banners, revenue curves, and best-seller charts.

CRITICAL AUTO-CHECKER & ALERT BANNERS

In the background, a validation engine constantly parses the dates and numbers. It triggers immediate alert flags in the main dashboard:

- Critical Expiry Banners: High alert (Red) if product expiry date is < 7 days away. Warning (Amber) if < 30 days.
- Critical Shortage Banners: Triggers when product stock levels drop below 3 units, preventing inventory stockouts.

3. Safe Database Manager & Sheets Integration

SPREADSHEET-STYLE DATA ENTRY & LOCK SAFETY

The "Manage Data" panel bridges direct cell manipulations with Google Sheets APIs. It is built to mimic the ease of standard spreadsheet editing while introducing data-entry validation checks and database collision safety locks:

- 1

Unsaved Changes Highlight - Double-clicking cells opens an inline input. Edited values are stored in a client state buffer (pendingChanges) and cells are highlighted in yellow until saved.
- 2

Transaction Collision Prevention - Enforces edit isolation. Operators are blocked from editing details on Row 6 if they have unsaved edits on Row 4, protecting sheets from overlapping save overwrites.
- 3

Dynamic Form Auto-completion - Adding a sale pre-populates unit descriptions and fetches active inventory selling prices, calculating transaction totals dynamically to prevent arithmetic errors.

PROJECT SCREENSHOT: DUAL-TAB DATA MANAGER & INLINE EDITING

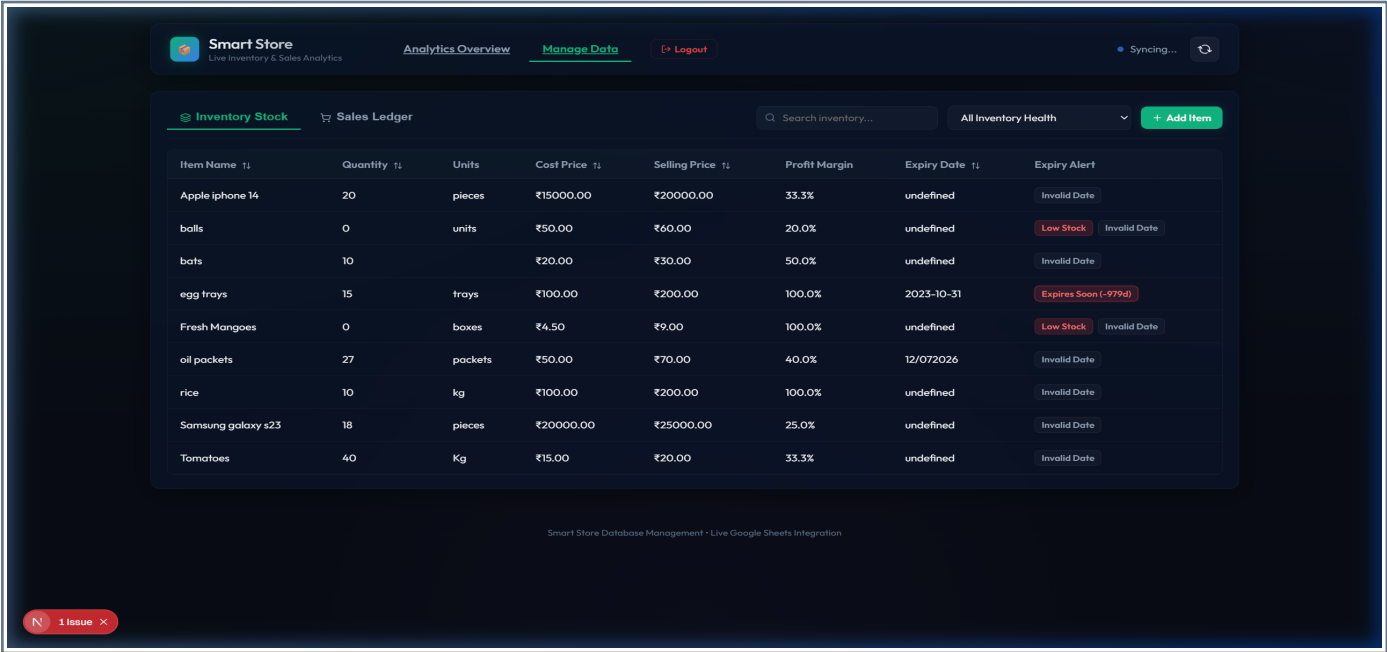


Figure 2: The Manage Data layout. Displays tab switching, live cell updates, sorting columns, search boxes, and adding rows.

AUTO-CACHE RESILIENCE & OFFLINE ROBUSTNESS

AuraRetail does not crash if the Google Sheets API limits are exceeded or if the shop's Wi-Fi drops out. The system instantly detects the offline state, shifts reading and writing to lib/mock_db.json, and fires a visual banner informing the admin that the application is running safely on its local offline cache.

4. Voice Assistant Pipeline & Integration Models

NATURAL SPEECH CONTEXT PIPELINE

The Voice Assistant widget provides hands-free operation. This is especially useful during busy store hours when shopkeepers have their hands full handling physical products. The audio workflow follows a strict, safe execution path:

- **1. Speech Capture** - Takes speech input from browser microphone using SpeechRecognition (Webkit Speech API) configured for en-IN accent mapping.
- **2. User Validation Check** - Transcribes and prints words on screen. Shopkeepers can inspect and edit the text first, clicking "Confirm & Send" to proceed.
- **3. n8n Workflow Gateway** - Sends text to n8n webhook proxy API. n8n acts as the AI router (linking LLMs like GPT or Claude) to parse data values.
- **4. Server-Side TTS Synthesis** - If n8n returns text, our backend proxies it via Google Translate TTS to render a binary audio stream, speaking the values back.

SPEECH & AUTOMATION MODELS INTEGRATION

- Speech-to-Text Model: Webkit Speech Recognition Engine (browser-native, lightweight, optimized for en-IN dialects).
- Text-to-Speech Model: Google Translate Text-To-Speech API (converts textual n8n returns under 200 characters into synthetic MP3 bytes).
- AI Agent workflow: Custom n8n workflow webhook proxy (for advanced cognitive routing, AI calculations, and sheet edits).

FAIL-SAFE LOCAL OFFLINE VOICE SHORTCUTS

If n8n is unconfigured or the server loses internet connectivity, the Voice Assistant falls back to local string matching to answer questions directly:

Voice Commands	Feature Focus	Local Action Response
"profit today", "net profit"	Finance KPIs	Speaks current profit margins (computed in context state)
"revenue today", "gross income"	Finance KPIs	Speaks current sales revenue total.
"stock value", "valuation"	Inventory valuation	Speaks the calculated total cost valuation of active stock
"low stock", "shortage alerts"	Safety threshold	Speaks count of items running below safe threshold levels.
"go to database", "open ledger"	Dashboard Navigation	Navigates active page route to the Database Manager.
"sync spreadsheet", "sync"	Manual Data Sync	Triggers background cell sync from Google Sheets API.

5. Technical Setup & Judges Project Pitch

PRODUCTION TECH STACK & SPECIFICATIONS

- Next.js App Router: Provides fast server-rendered API endpoints and responsive client components.
- Recharts: Renders beautiful area/bar charts with smooth transitions.
- Google Sheets API (v4): Accesses spreadsheet cells using JWT keys and OAuth client protocols.
- n8n Webhook: Serves as the cognitive gateway, forwarding transcripts to generative AI workflows.

ENVIRONMENT VARIABLES (.env.local)

```
GOOGLE_SERVICE_ACCOUNT_EMAIL=auraretail@ai-business-assistant-13998.iam.gserviceaccount.com
GOOGLE_PRIVATE_KEY="-----BEGIN PRIVATE KEY-----\nMIIIEvgIBADANBgkqhkiG9w0BAQEFA... \n-----END PRIVATE KEY"
GOOGLE_SPREADSHEET_ID=https://docs.google.com/spreadsheets/d/10Y008pUfV0VM3RVpCT.../edit
N8N_WEBHOOK_URL=https://hemanthmallela.app.n8n.cloud/webhook/f49f9beb-a9d2-4c01-8fc5...
USE MOCK DATA=false
ADMIN_USERNAME=admin
ADMIN_PASSWORD=admin123
```

WHY OUR PROJECT DESERVES THE HIGHEST RATING

1. Practical Social Impact: Small shops represent the backbone of retail economies. AuraRetail gives them enterprise-grade analytics tools with zero hardware costs, simply using a Google Sheet.
2. Advanced Voice AI Integration: We go beyond standard text inputs. Shopkeepers use voice commands to check profits or update sheets, while our server-side speech synthesis answers them verbally, making tech accessible to non-technical users.
3. Bulletproof Fail-safe Design: We do not fail. If API limits are exceeded or the network is cut, the application automatically shifts to a local mock DB JSON backup. Store data is saved locally and synced when online.
4. Complete UX Polish: The user experience is state of the art, utilizing dark modes, visual charts, safety locks, and immediate banners to handle errors gracefully.

CONCLUSION

AuraRetail successfully merges cloud databases, analytics dashboards, and interactive speech APIs into an offline-first retail operations tool. Ready for presentation.

